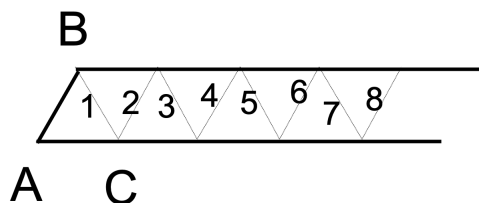


Problem Set 23

UMass Recreational Math Club

2025

Problem 1. A paper strip is divided into 2000 equilateral triangles along the dashed lines. The strip is successively folded at the lines in such a way that the strip always remains horizontal and the already folded part goes to the right. What is the position of triangle ABC after 1999 folds?



Problem 2. A polygon-shaped piece of paper is folded in half along a line of symmetry. The resulting shape is also folded in half, again along a line of symmetry. The final shape is a triangle. How many possibilities are there for the number of sides of the original polygon?

Problem 3. If a rectangular piece of paper is folded once, what is the smallest value of n (greater than 4) for which it is not possible to create an n -sided polygon in the same way?

Problem 4. A strip of paper is folded over the line through its center three times. It is then unfolded and viewed from the side, so that all seven resulting folds can be viewed at once. Which of the following is not a possible result?

